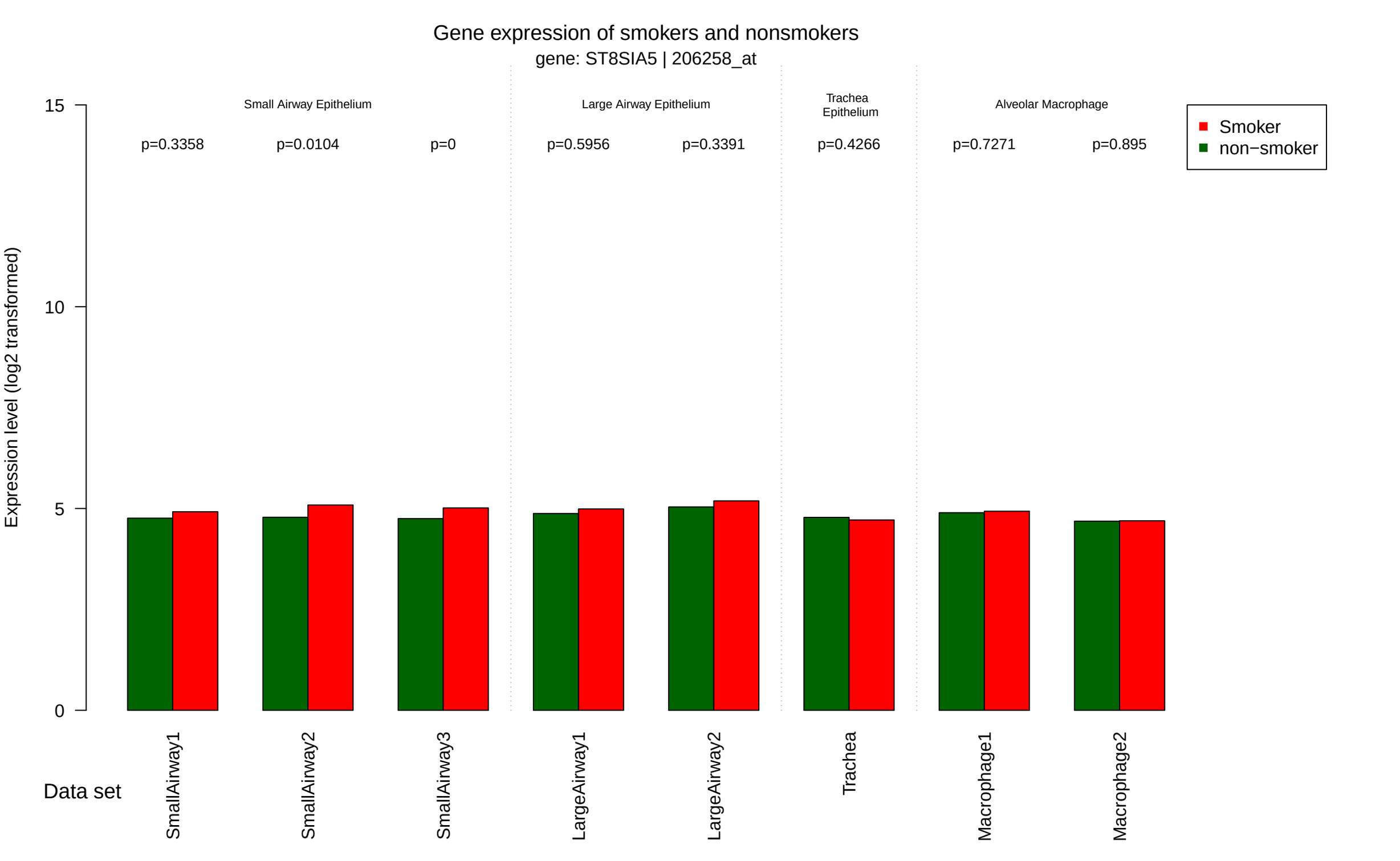
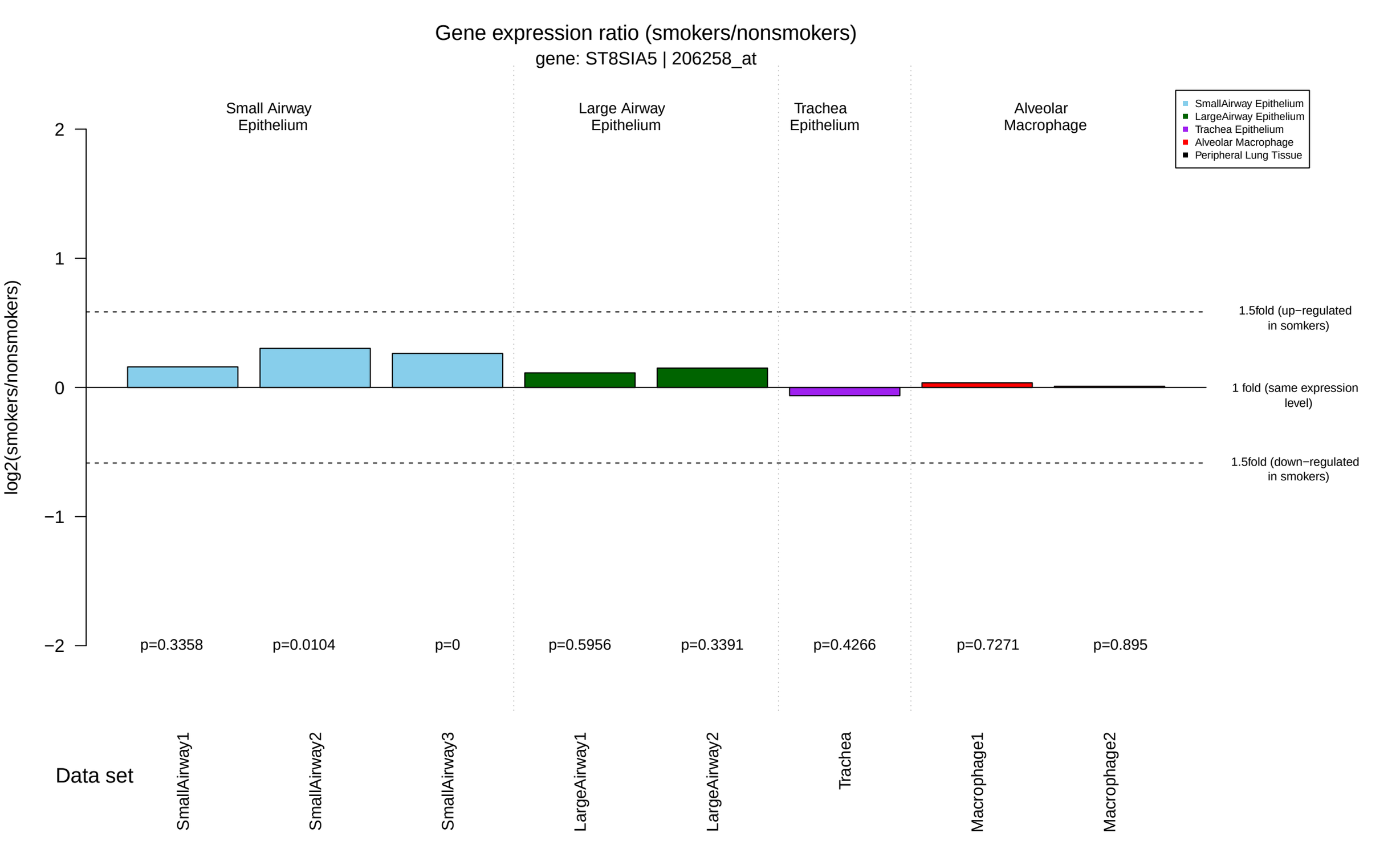
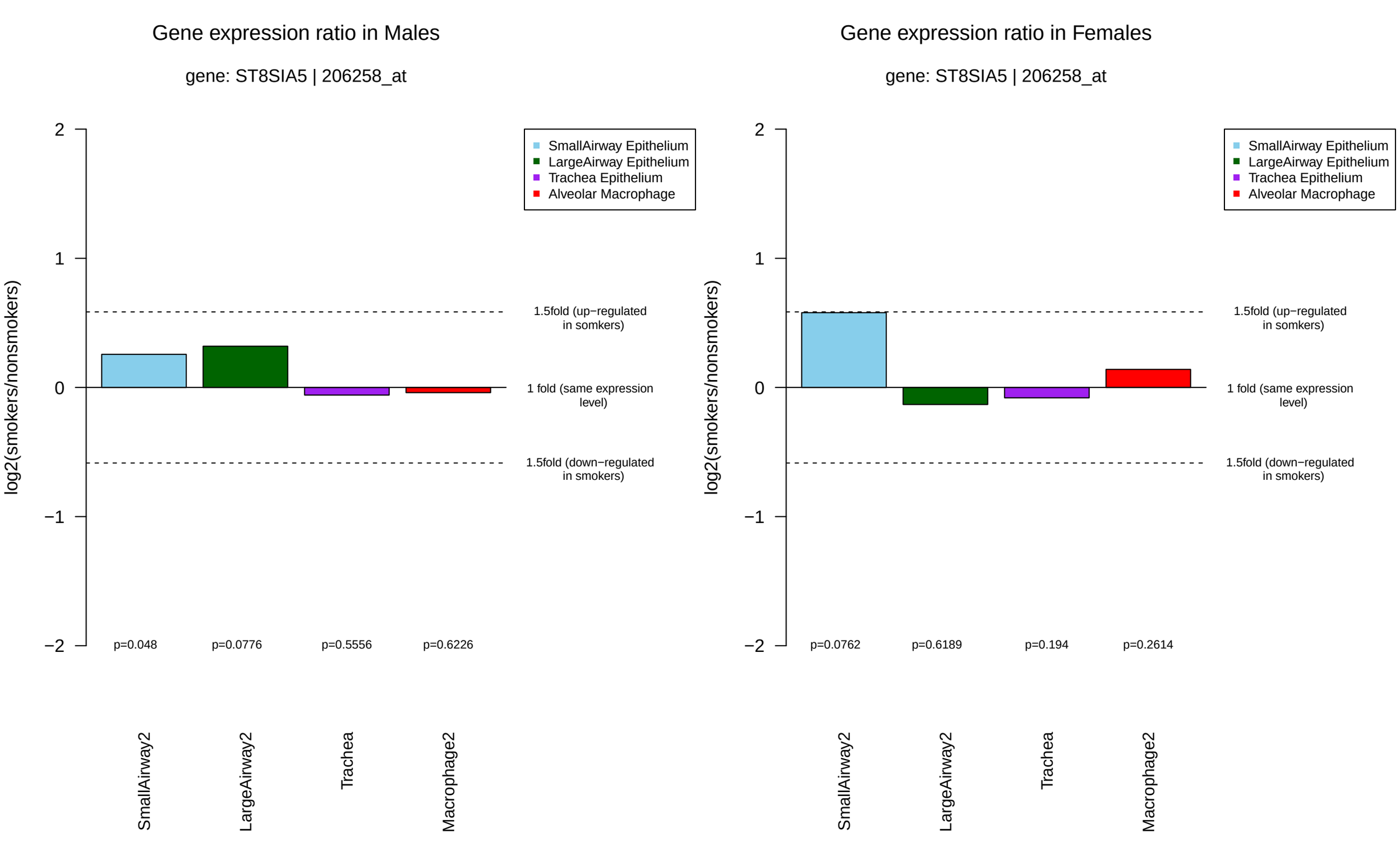
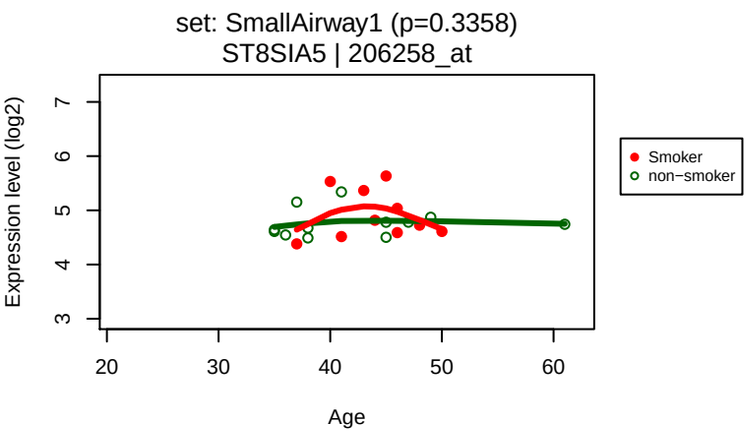
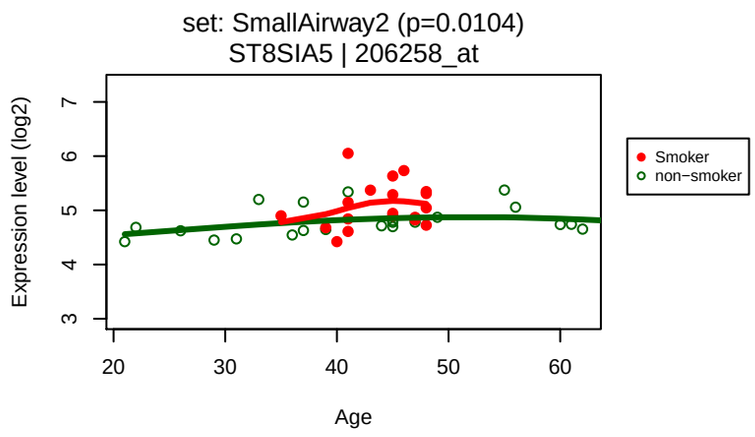
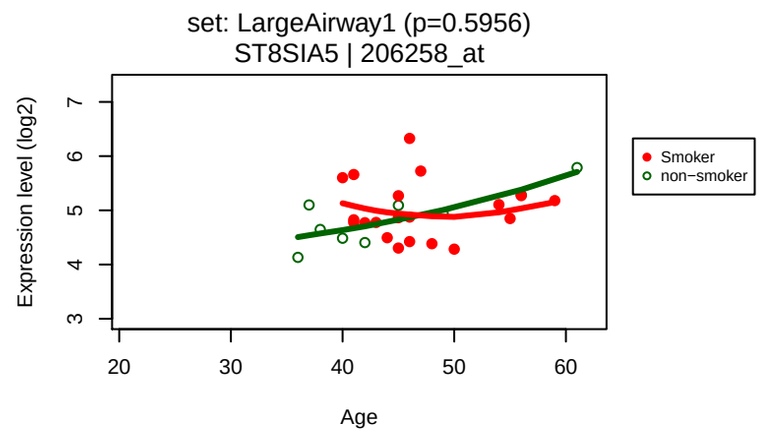

Gene expression of smokers and nonsmokers
gene: ST8SIA5 | 206258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3358
p=0.0104
p=0
p=0.5956
p=0.3391
p=0.4266
p=0.7271
p=0.895
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

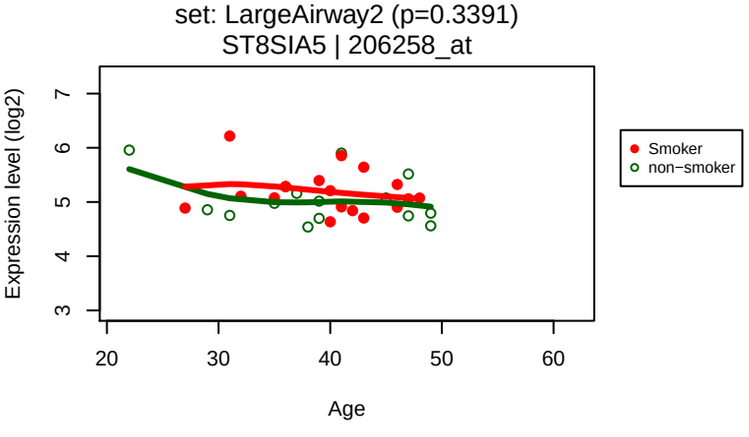

Gene expression ratio (smokers/nonsmokers)
gene: ST8SIA5 | 206258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3358
p=0.0104
p=0
p=0.5956
p=0.3391
p=0.4266
p=0.7271
p=0.895
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

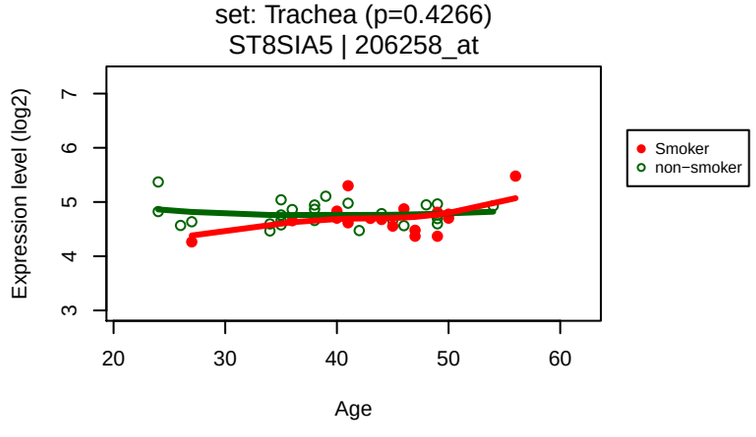

Gene expression ratio in Males
gene: ST8SIA5 | 206258_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.048
p=0.0776
p=0.5556
p=0.6226
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST8SIA5 | 206258_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0762
p=0.6189
p=0.194
p=0.2614
SmallAirway2
LargeAirway2
Trachea
Macrophage2

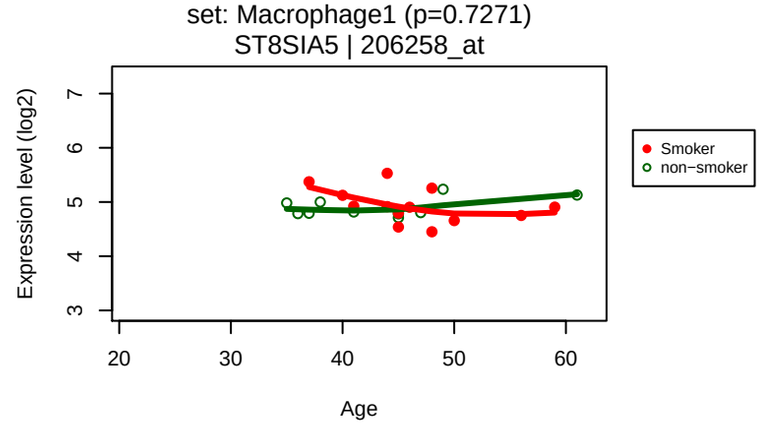

set: SmallAirway1 (p=0.3358)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker

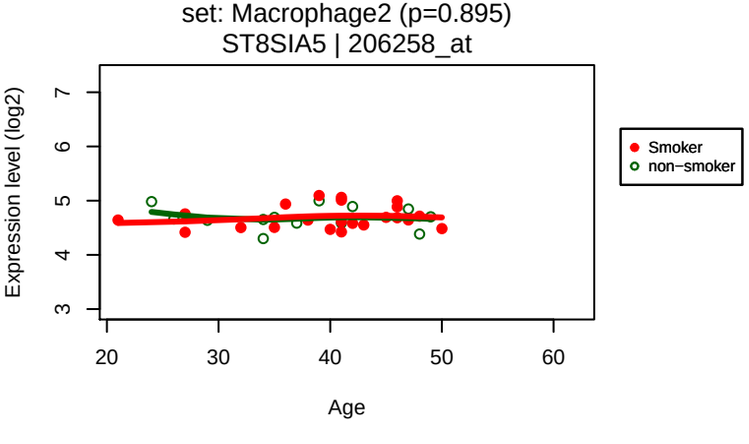

set: SmallAirway2 (p=0.0104)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker

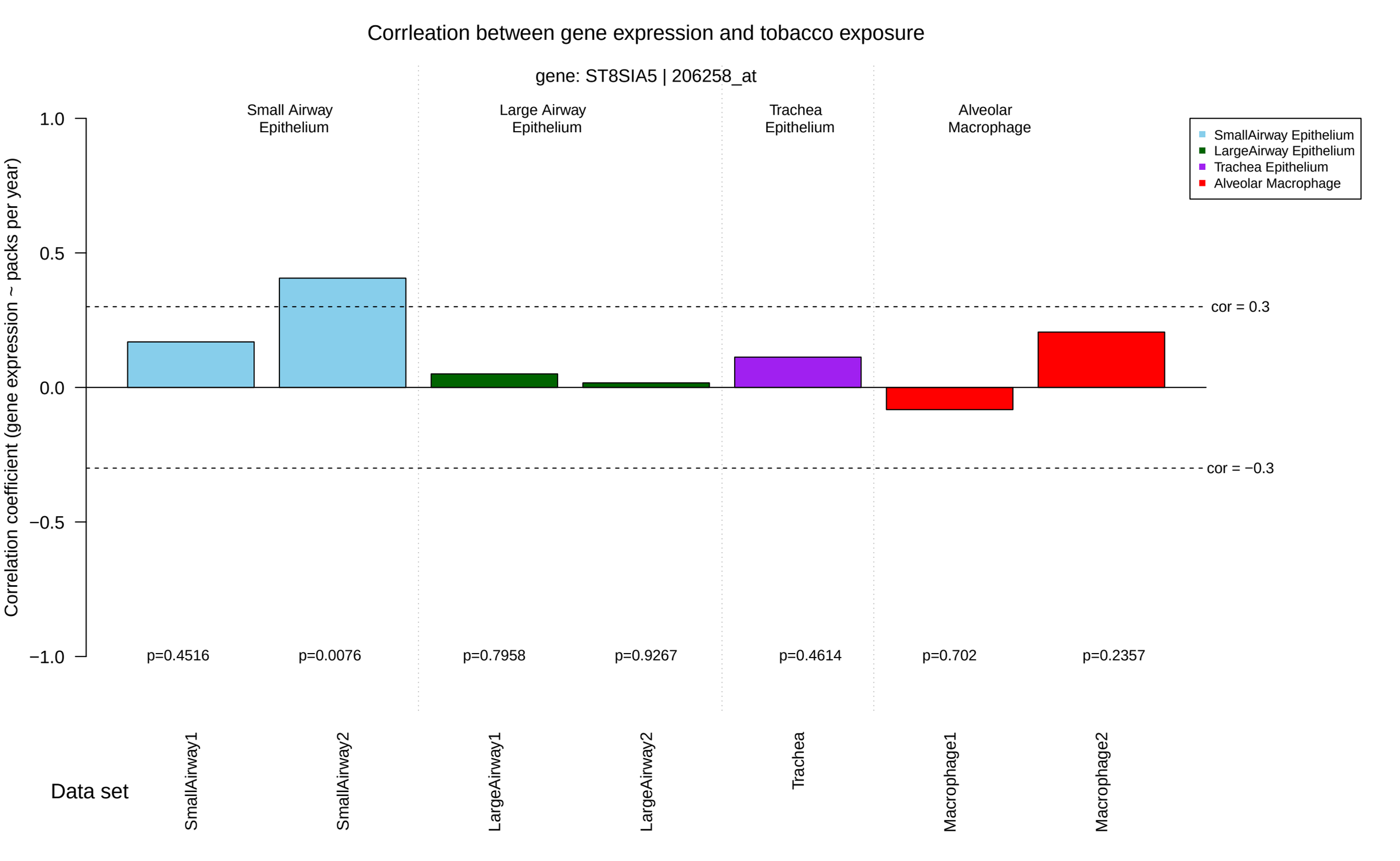

set: LargeAirway1 (p=0.5956)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3391)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4266)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7271)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.895)
ST8SIA5 | 206258_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST8SIA5 | 206258_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4516
p=0.0076
p=0.7958
p=0.9267
p=0.4614
p=0.702
p=0.2357
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2